

Identifying Rational Types in Unknown Environments under an Indirect Dynamic Mechanism

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Abstract

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This paper develops a dynamic indirect mechanism to identify rational types in unknown environments characterized by observational pooling. The central theoretical result, the Rational Type Identification Theorem, proves that the principal's Bayesian beliefs converge almost surely to the agent's true latent type along prolonged interactions without direct revelation reports. This asymptotic identification is operationalized by the Mano de Dios-Guadalupe process, an endogenous maximum-entropy noise kernel that ensures full support across actions within a competing-risks architecture. A complementary Implementability Theorem guarantees execution in Perfect Bayesian Equilibrium. Microeconomic calibration and deep-learning simulations of Colombian kidnapping validate rapid type separation and policy feasibility.

Keywords: indirect mechanism design, dynamic screening, endogenous implementation noise, Bayesian learning, type identification

JEL Classification: C73, D82, K42

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